M. Hesham

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SKILLS

Languages: Experienced in Python, C++, C. Familiar with C#, Java, HTML/CSS, JavaScript, SQL and MATLAB.

Tools & Platforms: FastAPI, AWS, Linux, Git/GitHub, Computer networks troubleshooting.

AI/ML: Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, LLM Serving (vLLM), Prompt Engineering.

EXPERIENCE

Texas State University, San Marcos, TX — Jan 2024 – Dec 2025
Graduate Assistant

- Improved department web presence for 70+ users by designing and developing an internal website using Python and Flask, resulting in a 30% reduction in staff response time to student inquiries.
- Reduced manual operational workload by 50% for a 15-person department by engineering Python automation scripts that eliminated repetitive administrative tasks, reclaiming 10+ staff hours per week and enabling focus on higher-priority initiatives.
- Reduced system failure response time by 40% by building a Python-based health monitoring service that polled 10+ Linux servers on configurable intervals, surfaced anomalies via automated alerts, and integrated with the department's existing incident workflow.

J&M Jewelry, San Marcos, TX — Jan 2025 – Apr 2025
Software Engineer Intern

- Improved order accuracy by 25% for 500+ monthly customers by engineering LLM-powered extraction pipelines in Python that parsed natural language chat into validated ring design parameters, enforcing deterministic output schemas and full audit logging.
- Reduced invalid configuration outputs by 30% by implementing 25+ validation rules in Python enforcing inventory constraints.

Orange Business, Cairo, Egypt — Dec 2019 – Dec 2023
Software Engineer

- Eliminated manual compliance review for 3 EMEA enterprise clients by deploying FastAPI microservices and REST APIs that automated data validation, reducing processing time by 70%.
- Reduced mean time to detection by 35% across 50+ enterprise environments by engineering a real-time Python ingestion pipeline that aggregated distributed performance metrics and surfaced them via REST APIs, enabling proactive incident response at scale.
- Reduced mean time to resolution by 40% by architecting an event-driven alerting system using RabbitMQ to detect real-time service degradations and auto-trigger remediation workflows, eliminating manual on-call escalation in 80% of incidents.

PROJECTS

RAG Document Assistant (Local LLM Pipeline) — **GitHub**

- Enabled natural language querying over private documents by engineering an end-to-end RAG pipeline using Python, FastAPI, LLaMA 3.1, and ChromaDB, delivering source-cited answers across 76+ indexed document chunks.
- Eliminated cloud API dependencies by serving LLaMA 3.1 8B locally via Ollama with 4-bit quantization, enabling fully offline inference on consumer hardware with zero cost per query.
- Reduced hallucinated LLM outputs by constraining generation to semantically retrieved context using 384-dimensional vector embeddings, ensuring grounded responses tied to source documents.
- Automated document ingestion by building a modular pipeline using LangChain that recursively loads, chunks, and embeds PDFs and text files into a persistent vector store with configurable chunk sizing and overlap.

High-Performance Computing & Performance Optimization — **GitHub**

- Built distributed data-processing pipelines using Python and Dask across multi-node GPU clusters, designing for scale and fault tolerance consistent with HPC and AI infrastructure requirements.
- Profiled kernel execution metrics using performance analysis tooling to optimize scheduling strategies and runtime throughput, reducing bottlenecks across parallel workloads on multi-node compute environments.
- Implemented compiler-aware optimization workflows with structured performance reporting, enabling systematic platform characterization and benchmarking across iterations.

EDUCATION

Texas State University, San Marcos, TX — Dec 2025
Master of Science, Computer Science | Unity Scholarship Winner — GPA: 3.92

Arab Academy for Science, Technology, Cairo, Egypt — Jun 2019
Bachelor of Engineering, Electronics and Communications